

SMART INDIA HACKATHON 2026

TATPAR

तत्पर • “ever-ready”

Readiness Assurance Platform for Air Fleets

Others predict failures. TATPAR assures readiness.



AI prognostics • sustainment digital twin • fleet optimisation • offline & sovereign

PROBLEM STATEMENT ID

26249

PROBLEM STATEMENT TITLE

Air Power – Predictive Maintenance & Fleet Availability

ORGANISATION

Ministry of Defence • Defence Services Staff College

THEME

Transportation & Logistics

PS CATEGORY

Software

TEAM ID

To be filled

TEAM NAME

To be filled

Proposed Solution

TATPAR joins health monitoring, technical records, spares and repair-agency data, then plans flying, maintenance and spares so that the required number of aircraft is mission-capable on the day of need — with stated confidence.



“How many aircraft on day D — and how sure?”

Readiness Forecast

A Fleet Twin of aircraft, spares, BRD/HAL repair pipeline, bays and manpower runs 200 Monte-Carlo futures → P10–P90 aircraft per day and P(meet task).



“What must I do today to have N ready?”

Readiness-Backward Planner

Optimises flying hours per tail, check timing, spares transfers, depot priority and cannibalisation; shows each action’s readiness gain; a human approves.



“Where is readiness lost — and why?”

Readiness-Loss Waterfall

Aircraft-days lost to spares, scheduled and unscheduled work, depot and manpower, with the days each lever could recover.

What is unique

Twin of the sustainment system — not a 3D model of one aircraft

Calibrated RUL — conformal 90 % intervals + engine-module attribution

Prognostics-aware flight & maintenance planning — staggered checks, bundled repairs

Multi-echelon spares (VARI-METRIC) fed by predicted demand

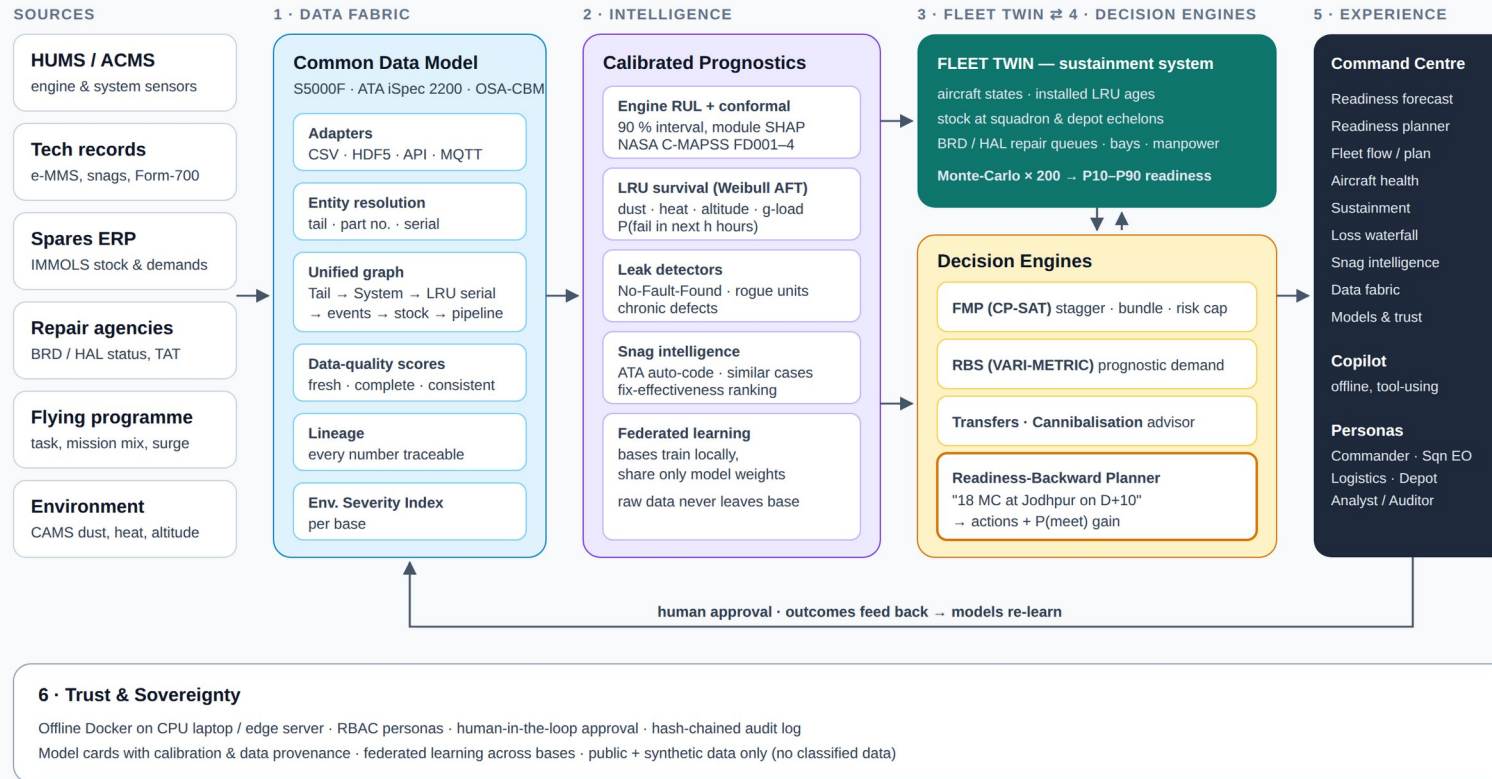
No-Fault-Found, rogue-unit & cannibalisation analytics

Indian base environment severity from real dust data; offline & federated

Technical Approach

TATPAR — Readiness Assurance Architecture

Integrate → Predict (calibrated) → Simulate (Fleet Twin) → Optimise → Assure → Learn · runs fully offline



Technology stack

Prognostics

LightGBM quantile + conformal (CQR), SHAP, Weibull AFT (lifelines)

Twin & optimisation

NumPy Monte-Carlo Fleet Twin, OR-Tools CP-SAT, VARI-METRIC

Platform

FastAPI · React + TypeScript · ECharts · SQLite → Postgres/TimescaleDB

Data

NASA C-MAPSS, MaintNet logbooks, CAMS dust, notional synthetic fleet

Deploy

Docker, fully offline on CPU laptop / edge server

Integrate

Predict

Simulate

Optimise

Approve

Learn

Feasibility & Viability

+20 pts

mission-capable in the working prototype's Fleet Twin: 57.7 → 78.0 %, same fleet and spares budget (notional data)

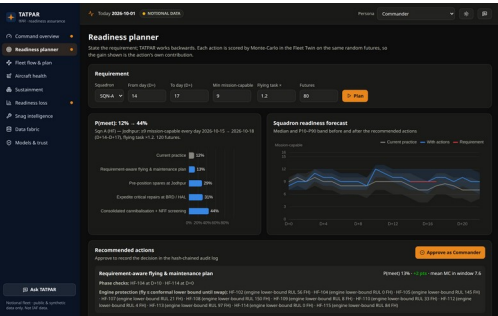
90 %

calibrated engine-RUL interval coverage on NASA C-MAPSS test sets (82.7 % before conformal calibration)

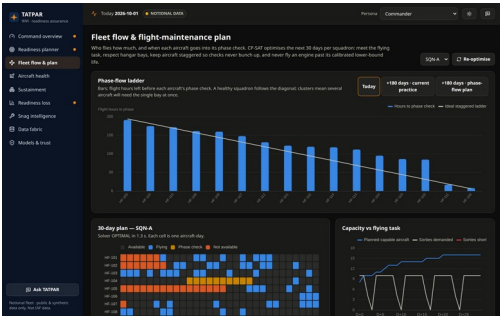
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cloud dependencies — runs air-gapped on a CPU laptop or edge server; public + synthetic data only

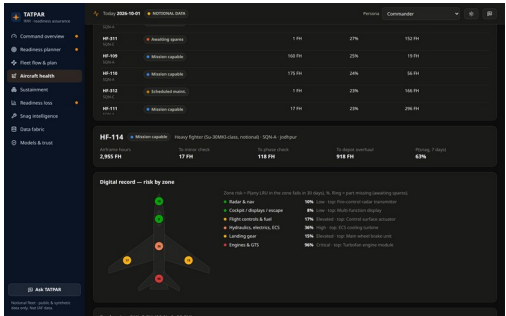
Challenge / risk	Mitigation built into TATPAR
No access to real IAF data	Public datasets + synthetic fleet with hidden ground truth; adapters for IMMOLS / e-MMS exports
Trust in AI recommendations	Calibrated intervals, explanations, human approval, hash-chained audit log
Classified, air-gapped networks	Fully offline; federated learning shares only model weights
Legacy and paper records	Common data model (S5000F / ATA), CSV adapters, data-quality scores
Optimisation run-time	CP-SAT plans a squadron in ~1–2 s; rolling horizon, per-squadron decomposition



Readiness planner



CP-SAT flight & maintenance plan



Engine RUL + module attribution

Working prototype built. Path to deployment: SIH prototype → pilot on one squadron's historical exports → BRD + IMMOLS integration → command-wide rollout.

Impact & Benefits

~40 %

of fighters reported unserviceable at any time
(post-Op Sindoor commentary)

20–70 %

of military avionics removals end as No-Fault-Found — wasted spares and repair slots

≈ 13

more aircraft mission-capable every day from a 64-aircraft fleet in the prototype twin — same spares budget (notional data)

Who benefits



Commander

N aircraft on day D, with a stated confidence



Squadron Engineering Officer

Which tails fly, which go in, what to bundle



Logistics Officer

What to stock, move or expedite — and where



BRD / Depot

Which repairs add the most readiness first



Operational

More aircraft from the same fleet; surge plans with known odds



Economic

Spares bought for availability per rupee; fewer NFF and repeat repairs



Strategic

Sovereign, offline software; supply-risk visibility for indigenisation



Safety & people

Fewer cannibalisations; chronic defects caught early

Research & References

IAF context

- Business Standard (2014) — Su-30MKI serviceability 48–55 %
- CAPS Issue Brief 08/25 — AI in IAF logistics, IMMOLS, e-MMS
- The Week (Aug 2026) — ML for Sukhoi and Tejas fleets, edge AI
- Defense Mirror — 11 BRD Ojhar Su-30MKI overhaul

Operations research

- Kozanidis — Flight & maintenance planning, Hellenic AF
- Peschiera et al. — Long-term FMP, French AF (arXiv 2001.09856)
- Mattila & Virtanen (2014) — Fighter maintenance sim-optimisation
- Sherbrooke — VARI-METRIC, Operations Research (1986)
- GAO-02-86 — Military aircraft cannibalisations

Prognostics & AI

- Saxena et al. (2008) — NASA C-MAPSS turbofan dataset
- LSTM quantile + conformal RUL — Sensors 26(7) 2249 (2026)
- Federated RUL prognostics — arXiv 2506.00499
- MaintNet aviation logbooks — COLING 2020
- DST-Group-TR-3367 — dust-driven engine degradation

Gap analysis: 7 public PS-26249 projects reviewed (NIRANTAR, AERO-READY, AeroTwin-AI, AirPower, PREDIX, VAYU SEWA) — none model flight & maintenance planning, multi-echelon sparing or probabilistic readiness. Prototype figures use a notional fleet; no classified data.